

A BAYESIAN PROOF OF THE BERNOULLI THEOREM

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ABSTRACT. We give a new proof of the Bernoulli theorem, conjectured by Talagrand and proved in the seminal work of Bednorz and Latała. Our approach is based on information-theoretic ideas: lower bounds on the supremum of a Bernoulli process are translated to the fundamental limits of Bayesian estimation in a Cauchy additive channel. This leads to a new information-theoretic functional that characterizes Bernoulli-process suprema and plays a role analogous to Fernique’s majorizing-measure functional for Gaussian processes. The same viewpoint yields a distributional strengthening: for any prescribed law of the index, we characterize the largest expected value attainable over all couplings of that index with the Bernoulli process. This extends to Bernoulli processes a phenomenon previously understood for Gaussian processes through the work of Fernique and Talagrand.

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1. INTRODUCTION

Let $T \subset \mathbb{R}^n$ be finite. The celebrated majorizing measure theorem (MMT), whose upper bound is due to Fernique and whose matching lower bound was proved by Talagrand [Fer75, Tal87], gives a sharp characterization, up to universal constants, of the Gaussian width of T ,

$$W(T) := \mathbb{E} \sup_{x \in T} \langle G, x \rangle, \quad G \sim N(0, I_n). \quad (1)$$

Write $\mathcal{P}(T)$ for the set of probability measures on T , and set

$$B_2(t, r) := \{s \in T : \|s - t\|_2 \leq r\}.$$

The MMT may then be stated as follows.

Theorem 1.1 (Majorizing Measure Theorem). *There are universal constants $c, C > 0$ such that every finite $T \subset \mathbb{R}^n$ satisfies*

$$c \gamma_2(T, \|\cdot\|_2) \leq W(T) \leq C \gamma_2(T, \|\cdot\|_2), \quad (2)$$

where

$$\gamma_2(T, \|\cdot\|_2) := \inf_{\mu \in \mathcal{P}(T)} \sup_{t \in T} \int_0^\infty \sqrt{\log \frac{1}{\mu(B_2(t, r))}} dr. \quad (3)$$

Beyond Gaussian processes, another central problem is to control suprema of Bernoulli processes, which arise naturally through symmetrization in empirical-process theory; see, for example, [Tal94, Tal21]. For a finite $T \subset \mathbb{R}^n$, define the Bernoulli width

$$b(T) := \mathbb{E} \sup_{x \in T} \langle \varepsilon, x \rangle, \quad (4)$$

where $\varepsilon_1, \dots, \varepsilon_n$ are independent centered Rademacher variables. Since ε is a subgaussian vector, the standard arguments underlying the upper bound in the MMT give $b(T) \leq C \gamma_2(T, \|\cdot\|_2)$. Moreover, since $\|\varepsilon\|_\infty = 1$ almost surely, $b(T) \leq \sup_{x \in T} \|x\|_1$. Talagrand's celebrated Bernoulli conjecture [Tal94] asserted that the optimal interpolation between these two elementary bounds is sharp for every finite T . The conjecture was resolved roughly two decades later in the breakthrough work of Bednorz and Latała [BL14]. Define

$$\Lambda(T) := \inf_{\substack{T_1, T_2 \subset \mathbb{R}^n \text{ finite} \\ T \subseteq T_1 + T_2}} \left\{ \sup_{u \in T_1} \|u\|_1 + \gamma_2(T_2, \|\cdot\|_2) \right\}. \quad (5)$$

The Bernoulli theorem of Bednorz and Latała can then be stated as follows.

Theorem 1.2 (Bernoulli theorem). *There are universal constants $c, C > 0$ such that every finite $T \subset \mathbb{R}^n$ satisfies*

$$c \Lambda(T) \leq b(T) \leq C \Lambda(T). \quad (6)$$

The upper bound follows immediately by combining the two elementary bounds above. The central challenge, and hence the principal focus of [BL14], is to prove the lower bound. Because the proof of Bednorz and Latała is technically involved, there has been considerable interest in finding a more direct and intuitive argument; see, for example, [Tal24].

The situation for the Gaussian MMT is now markedly different. Although Talagrand’s original proof was also considered opaque, several comparatively simple proofs are now available, including geometric arguments and approaches based on coding theory, interpolation, type lifting, functional analysis, and Bayesian estimation [Tal92, Tal96, vH18, BDOS21, Liu25, CR26, Zad26]. Most relevant here are two recent developments of the authors [Liu25, Zad26]. The first author showed in [Liu25] that the majorizing-measure functional $\gamma_2(T, \|\cdot\|_2)$ admits an equivalent information-theoretic formulation in terms of a rate–distortion integral, obtained naturally by a lifting argument. The second author subsequently showed in [Zad26] that this formulation yields an independent information-theoretic proof of the majorizing measure theorem. That proof expresses the Gaussian width through a Bayesian estimation problem for a Gaussian additive channel and then applies information-theoretic inequalities to obtain the lower bound.

In the present work, we show that the Bayesian approach also tightly characterizes the Bernoulli width. The central new observation is that the appropriate analogue of the Gaussian additive channel is a *Cauchy additive channel*. We adapt both the information-theoretic approach of [Liu25] and the Bayesian estimation perspective of [Zad26] to the Bernoulli width and the Cauchy channel. This yields a new and comparatively direct proof of the Bernoulli theorem.

Once the necessary ingredients have been introduced, the proof closely parallels the Bayesian proof of the Gaussian MMT in [Zad26], together with an elementary decomposition argument. By contrast, the Bednorz–Latała proof uses ingredients that do not arise in the Gaussian setting, including Latała’s principle for dropping coordinates, chaining functionals defined through chopping maps, and an adaptive decomposition procedure. Our approach bypasses this machinery. One additional ingredient is needed for our proof: the Cauchy-channel information–estimation inequality proved here, which plays the role of the Gaussian I–MMSE identity of [GSV05].

1.1. The distributional Bernoulli theorem. As the functional $\Lambda(T)$ in Theorem 1.2 yields a trivial upper bound on $b(T)$, the classical formulation of the Bernoulli theorem is conceptually appealing. At the heart of our approach, however, lies a new formulation of the Bernoulli theorem that characterizes $b(T)$ in terms an information-theoretic functional (eq. (10) below) that is the analogue in the present setting of the Gaussian functional introduced in [Liu25]. This new formulation not only provides a novel viewpoint on the Bernoulli theorem, but also enables us to establish a significantly stronger result.

Let Rad denote the uniform law on $\{-1, 1\}$, and define, for every finitely supported probability measure μ on $\mathbb{R}^n$,

$$B(\mu) := \sup_{\substack{X \sim \mu, \varepsilon \sim \text{Rad}^{\otimes n} \\ \text{jointly coupled}}} \mathbb{E}\langle \varepsilon, X \rangle. \quad (7)$$

That is, $B(\mu)$ is the largest possible expectation of the Bernoulli process evaluated at a random point X with distribution μ . The main result of this paper, Theorem 2.1 below, yields a distributional strengthening of the Bernoulli theorem: it provides a characterization of $B(\mu)$ for each μ individually. Since

$$b(T) = \sup_{\mu \in \mathcal{P}(T)} B(\mu),$$

Theorem 1.2 will follow as a corollary by taking the supremum over μ .

Fixed-law functionals analogous to $B(\mu)$ were introduced in the Gaussian setting by Fernique, who established several bounds for them [Fer81, Theorem 3.1]. Talagrand subsequently obtained universal-constant characterizations of the Gaussian fixed-law functional [Tal87, Theorem 30]; see also [vH25]. By contrast, the methods of Bednorz–Łatała do not appear to yield a corresponding fixed-law version of the Bernoulli theorem. Thus, to our knowledge, our distributional Bernoulli theorem is the first such fixed-law characterization for a non-Gaussian process.

1.2. Organization of the paper. Section 2 introduces the information-theoretic objects and states the main distributional theorem. Sections 3 and 4 compare the Bernoulli width with the integrated Cauchy Bayes risk and then with the rate–distortion functional. Section 5 converts this functional into the ℓ_1 -plus-Gaussian decomposition, and Section 6 proves the prescribed-law distributional result. The Cauchy information–estimation inequality is proved in Appendix A.

Statement on AI use. ChatGPT Pro (versions 5.5 and 5.6) was used to generate an initial proof sketch for the information-estimation inequality in Appendix A, specifically in response to the authors’ query regarding whether the inequality holds for the Cauchy additive channel. Beyond this, ChatGPT Pro (versions 5.5 and 5.6) was utilized for accelerating technical calculations (specifically in the proofs of Lemma 3.6 and Lemma 5.3) and language editing throughout the paper. The authors take full responsibility for all content and its accuracy.

Acknowledgments. We thank Ramon van Handel for sharing unpublished notes, which served as a major motivation for the rate–distortion integral characterization of Bernoulli processes described in Theorem 2.1. In particular, these notes establish a Dudley-type integral bound for sets of sequences with a prescribed empirical distribution, known as exact type classes; together with the Bernoulli theorem as input, this implies the rate–distortion characterization. In the present paper, we instead use the first part of our new Bayesian proof (specifically Eq. (18)) to obtain the rate–distortion characterization, which in turn yields an independent proof of the Bernoulli theorem. We are also deeply grateful to him for several insightful discussions and for his helpful feedback on early versions of this manuscript.

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2. MAIN RESULT AND PROOF STRUCTURE

2.1. Notation. All logarithms are natural and $0 \log 0 = 0$. For $A, B \geq 0$, $A \lesssim B$ means $A \leq CB$ for a universal constant $C > 0$, $A \gtrsim B$ means $B \lesssim A$, and $A \asymp B$ means both.

For probability measures P, Q , set

$$D_{\text{KL}}(P \| Q) := \begin{cases} \int \log \left(\frac{dP}{dQ} \right) dP, & P \ll Q, \\ +\infty, & P \not\ll Q. \end{cases}$$

For random variables X, Y, S , we define

$$I(X; Y) := D_{\text{KL}}(P_{X,Y} \| P_X \otimes P_Y)$$

and

$$I(X; Y | S) := \int D_{\text{KL}}(P_{X,Y|S=s} \| P_{X|S=s} \otimes P_{Y|S=s}) P_S(ds).$$

These definitions apply equally to discrete and continuous observations. If X is discrete, then

$$H(X) := - \sum_x \mathbf{P}(X = x) \log \mathbf{P}(X = x)$$

and

$$H(X | Y) := \mathbb{E} \left[- \sum_x \mathbf{P}(X = x | Y) \log \mathbf{P}(X = x | Y) \right], \quad I(X; Y) = H(X) - H(X | Y).$$

For a probability measure μ on a finite set T , we also write

$$H(\mu) := - \sum_{x \in T} \mu(x) \log \mu(x).$$

For $v : T \rightarrow \mathbb{R}^n$, write $v_{\#}\mu$ for the pushforward of μ under v . We use without further comment nonnegativity, the chain rule

$$I(X; Y, Z) = I(X; Y) + I(X; Z | Y),$$

and the data-processing inequality: if $X \rightarrow Y \rightarrow Z$ is a Markov chain, then $I(X; Z) \leq I(X; Y)$.

We will also use posterior replicas. Given two random variables X, Y where Y is a (random) function of X (in statistical terminology, Y is often called an observation of the signal X), a posterior replica $X^{(1)}$ is a random variable with conditional law $P_{X|Y}$. Two posterior replicas $X^{(1)}, X^{(2)}$ are sampled conditionally independently of each other and of X given Y . Thus $X, X^{(1)}, X^{(2)}$ are conditionally i.i.d. given Y , and in particular

$$\mathbb{E} f(X, X^{(1)}) = \mathbb{E} f(X^{(1)}, X^{(2)})$$

for every integrable measurable f . We refer to this elementary posterior-sampling identity as the Nishimori identity.

2.2. **Key objects.** For $t > 0$ and $x, x' \in \mathbb{R}^n$, define the capped quadratic distortion

$$\varphi_t(x, x') := \sum_{i=1}^n \left(1 \wedge \frac{|x_i - x'_i|^2}{t^2} \right). \quad (8)$$

This distortion already appears in Talagrand's geometric formulation of the Bernoulli problem [Tal94, Eq. (4.4)]. It captures the scale-dependent quadratic geometry that contributes to the Gaussian component of the Bernoulli decomposition. It also motivates our use of the Cauchy channel: the Cauchy information–estimation inequality proved below naturally produces precisely the capped loss φ_t .

Fix a finite $T \subset \mathbb{R}^n$. For $1 \leq p \leq \infty$, write

$$\text{diam}_p(T) := \max_{x, u \in T} \|x - u\|_p.$$

For $\mu \in \mathcal{P}(T)$, define

$$\text{RD}_\mu(t) := \inf_{X \sim \mu, \widehat{X} \sim \mu} \left\{ I(X; \widehat{X}) + \mathbb{E} \varphi_t(X, \widehat{X}) \right\}, \quad (9)$$

where the infimum is over all couplings with both marginals equal to μ . Thus $\text{RD}_\mu(t)$ is a self-coupled Lagrangian rate–distortion value. The set-level rate–distortion functional is

$$R(T) := \sup_{\mu \in \mathcal{P}(T)} \int_0^\infty \text{RD}_\mu(t) dt. \quad (10)$$

We next define the Cauchy channel and its Bayes risk. Let $Z = (Z_1, \dots, Z_n)$ have independent standard Cauchy coordinates, each with density

$$z \mapsto \frac{1}{\pi(1 + z^2)}.$$

For $\mu \in \mathcal{P}(T)$, let $X \sim \mu$ be independent of Z , and observe

$$Y_t := X + tZ. \quad (11)$$

The corresponding capped minimum mean-square error is

$$\text{cMMSE}_\mu(t) := \inf_{\widehat{x}: \mathbb{R}^n \rightarrow \mathbb{R}^n} \mathbb{E} \varphi_t(X, \widehat{x}(Y_t)), \quad (12)$$

where the infimum is over measurable estimators, which need not be T -valued.

For every finitely supported probability measure μ on $\mathbb{R}^n$, define the prescribed-law Gaussian functional

$$G(\mu) := \sup_{\substack{Y \sim \mu, G_0 \sim N(0, I_n) \\ \text{jointly coupled}}} \mathbb{E} \langle Y, G_0 \rangle.$$

Finally, with $\text{id}_T(x) = x$, set

$$\delta_T(\mu) := \inf_{a: T \rightarrow \mathbb{R}^n} \left\{ \mathbb{E}_{X \sim \mu} \|a(X)\|_1 + G((\text{id}_T - a)_\# \mu) \right\}. \quad (13)$$

2.3. Main result.

Theorem 2.1 (Distributional Bernoulli rate–distortion theorem). *For every finite $T \subset \mathbb{R}^n$ and every $\mu \in \mathcal{P}(T)$,*

$$B(\mu) \asymp \int_0^\infty \text{RD}_\mu(t) dt \asymp \delta_T(\mu), \quad (14)$$

where the implicit constants are universal. Consequently,

$$b(T) \asymp R(T) \asymp \Lambda(T). \quad (15)$$

The prescribed-law statement (14) refines the classical Bernoulli theorem by characterizing the largest expected Bernoulli-process value attainable over all couplings for which the index has a prescribed law. An analogous fixed-law characterization is available for Gaussian processes [Tal87, Theorem 30], whereas existing proofs of the Bernoulli theorem do not appear to yield such a prescribed-law result. Furthermore, the proof of (14) reveals that the decomposition in δ_T (function a in (13)) can be constructed using information-constrained near-optimal transport couplings. Supremizing μ over T in (14) gives (15).

2.4. Proof structure and ideas. The proof of Theorem 2.1 closes a cycle of three comparisons. Type lifting [Liu25], which replaces an index law by the uniform law on sequences with that empirical distribution together with a comparison with the fundamental Bayesian estimation limits of a Cauchy channel, gives $\int_0^\infty \text{RD}_\mu(t) dt \lesssim B(\mu)$, an elementary coupling gives $B(\mu) \lesssim \delta_T(\mu)$, and a multiscale distributional decomposition gives $\delta_T(\mu) \lesssim \int_0^\infty \text{RD}_\mu(t) dt$. Taking the supremum over the μ and applying the corresponding minimax decomposition yields the set-level assertion.

Here is how the three comparisons above are distributed through the paper. Sections 3 and 4 first prove Theorem 2.2, and hence $\int_0^\infty \text{RD}_\mu(t) dt \lesssim b(T)$ for every μ . Section 5 proves $\delta_T(\mu) \lesssim \int_0^\infty \text{RD}_\mu(t) dt$ and uses a minimax argument to prove Theorem 2.3. Finally, Section 6 applies the set-level estimate to uniform laws on empirical-distribution fibers to obtain $\int_0^\infty \text{RD}_\mu(t) dt \lesssim B(\mu)$, proves $B(\mu) \lesssim \delta_T(\mu)$ by Gaussianization, and completes the prescribed-law part of Theorem 2.1.

At the set level, Theorem 2.2 gives $R(T) \lesssim b(T)$, Theorem 2.3 gives $\Lambda(T) \lesssim R(T)$, and the elementary ℓ_1 and Gaussian bounds applied to a decomposition $T \subseteq T_1 + T_2$ give $b(T) \lesssim \Lambda(T)$. The two non-elementary set-level estimates are stated next.

Theorem 2.2 (Bernoulli width dominates the rate–distortion area). *For every finite $T \subset \mathbb{R}^n$ and every $\mu \in \mathcal{P}(T)$,*

$$b(T) \geq \frac{1}{4\pi} \int_0^\infty \text{cMMSE}_\mu(t) dt, \quad (16)$$

$$\int_0^\infty \text{cMMSE}_\mu(t) dt \geq 0.05 \int_0^\infty \text{RD}_\mu(t) dt. \quad (17)$$

Consequently,

$$b(T) \geq \frac{1}{80\pi} R(T). \quad (18)$$

	Gaussian MMT	Bernoulli theorem
Additive channel	$Y_t^G = X + tG, G \sim N(0, I_n)$	$Y_t^C = X + tZ, Z_i$ i.i.d. standard Cauchy
Natural loss	$\ x - \hat{x}\ _2^2$	$\varphi_t(x, \hat{x})$
Width-to-risk step	Gaussian width controls integrated Gaussian MMSE [Zad26]	Bernoulli width controls integrated Cauchy cMMSE (Section 3)
Information–estimation step	Gaussian I–MMSE identity [GSV05]	Cauchy information–estimation inequality (Appendix A)
Risk-to-rate–distortion step	Integrated MMSE controls a rate–distortion functional [Zad26]	Integrated cMMSE controls the rate–distortion functional (Section 4)
Functional step	The rate–distortion dual controls $\gamma_2(T, \ \cdot\ _2)$ [Liu25]	$R(T)$ controls $\Lambda(T)$ (Section 5)

TABLE 1. Dictionary between the Bayesian proofs in the Gaussian and Bernoulli settings.

The Gaussian argument that motivates Theorem 2.2 can be summarized without assuming prior familiarity with information theory. In a Gaussian additive channel, the Gaussian width can be expressed as the area under the estimation-error curve of a least-squares estimator (see also [PZ26]). Replacing that estimator by the Bayes-optimal estimator can only decrease the error. The Gaussian mutual-information/minimum mean-square error (I–MMSE) identity [GSV05] then relates this Bayes risk to the decay of mutual information, which in turn controls a rate–distortion quantity.

For Bernoulli processes, an interpolation potential based on the least absolute deviation estimator in the Cauchy channel yields the first inequality in Theorem 2.2. Its endpoint identity is

$$\lim_{t \rightarrow \infty} \mathbb{E}[\|Y_t - X\|_1 - \|Y_t - \Pi_K(Y_t)\|_1] = b(T), \quad \Pi_K(y) \in \arg \min_{u \in \text{conv}(T)} \|y - u\|_1.$$

Differentiating the potential produces a capped estimation risk. The Cauchy information–estimation inequality proved in Appendix A, together with a posterior-replica coupling, gives the second inequality. Motivated by the question raised in [Ver23], we derive this Cauchy-channel relation for finite input distributions and obtain the required bound in terms of integrated capped Bayes risk.

Theorem 2.3 (Rate–distortion area dominates the Bernoulli functional). *There is a universal constant $c > 0$ such that every finite $T \subset \mathbb{R}^n$ satisfies*

$$R(T) \geq c \Lambda(T). \quad (19)$$

For a prescribed-law μ , we prove that the rate–distortion integral controls $\delta_T(\mu)$. A minimax argument then converts $\sup_\mu \delta_T(\mu)$ into a single decomposition of T with an ℓ_1 -bounded part

and a part of controlled Gaussian width. The Gaussian MMT turns the latter bound into γ_2 -control, proving [Theorem 2.3](#).

3. BERNOULLI WIDTH AND THE CAUCHY CMMSE AREA BOUND

The goal of this section is to prove (16), the first comparison in [Theorem 2.2](#). We construct an interpolation potential whose total variation is $b(T)$, show that its derivative controls the capped error of a leave-one-coordinate-out estimator, and then average over the prior and optimize over estimators.

3.1. Area theorem for Bernoulli width and absolute-deviation estimation. We first construct the interpolation potential and identify its endpoints, thereby expressing $b(T)$ as the integral of its derivative.

Consider the convex hull $K := \text{conv}(T)$ and define the ℓ_1 -distance to K by

$$d_K(y) := \inf_{u \in K} \|y - u\|_1. \quad (20)$$

Remark 3.1. It is useful to adopt an information-theoretic or statistical viewpoint. Fix a Borel measurable selector

$$\Pi_K(y) \in \arg \min_{u \in K} \|y - u\|_1, \quad y \in \mathbb{R}^n.$$

Such a selector exists because K is compact and the objective is jointly continuous. In the Cauchy channel (11), $\Pi_K(Y_t)$ is a least absolute deviation estimator and

$$d_K(y) = \|\Pi_K(y) - y\|_1.$$

Now, fix a “planted” point $x \in T$. For $t \geq 0$, define the interpolation potential

$$\Phi_x(t) := \mathbb{E}_Z [t \|Z\|_1 - d_K(x + tZ)]. \quad (21)$$

Equivalently,

$$\Phi_x(t) = \mathbb{E}_Z \sup_{u \in K} \{t \|Z\|_1 - \|x + tZ - u\|_1\}. \quad (22)$$

Remark 3.2. Notice that although $\mathbb{E} \|Z\|_1 = \infty$, $\Phi_x(t)$ is well-defined. Indeed, for every $z \in \mathbb{R}^n$, $t \geq 0$, and $x \in T$, choosing $u = x$ in [Eq. \(22\)](#) for the lower bound and using the triangle inequality for the upper bound give

$$0 \leq t \|z\|_1 - d_K(x + tz) \leq \sup_{u \in K} \|u - x\|_1. \quad (23)$$

The first key lemma connects the Bernoulli width $b(T)$ with $\Phi_x(t)$.

Lemma 3.3 (Cauchy-area theorem for Bernoulli width). *For any $x \in T$, the function $\Phi_x : [0, \infty) \rightarrow [0, +\infty)$ is nondecreasing and concave on $[0, \infty)$ with $\Phi_x(0) = 0$. Moreover, it satisfies*

$$b(T) = \lim_{t \rightarrow \infty} \Phi_x(t) = \int_0^\infty \Phi'_x(t) dt. \quad (24)$$

The derivative is understood almost everywhere.

Proof. For $z \in \mathbb{R}^n$, set

$$F_z(t) := t\|z\|_1 - d_K(x + tz), \quad t \geq 0.$$

Clearly, $F_z(0) = -d_K(x) = 0$. Since d_K is convex, F_z is concave. Moreover, if $0 \leq t_2 \leq t_1$, the 1-Lipschitz property of d_K with respect to $\|\cdot\|_1$ gives

$$d_K(x + t_1 z) - d_K(x + t_2 z) \leq (t_1 - t_2)\|z\|_1,$$

and hence $F_z(t_1) \geq F_z(t_2)$. Taking expectations shows that Φ_x is concave and nondecreasing.

Let

$$D_x := \sup_{u \in K} \|u - x\|_1 < \infty.$$

By Eq. (23), $0 \leq F_z(t) \leq D_x$ for every z and $t \geq 0$. For every fixed z , $F_z(t) \rightarrow 0$ as $t \downarrow 0$. Dominated convergence therefore gives $\Phi_x(t) \rightarrow 0 = \Phi_x(0)$ as $t \downarrow 0$.

Now fix z such that $z_i \neq 0$ for every i . Compactness of K implies, for every i ,

$$\lim_{t \rightarrow \infty} \sup_{u \in K} \|x_i + tz_i - u_i\| - t|z_i| - \text{sgn}(z_i)(x_i - u_i) = 0.$$

Indeed, when $z_i > 0$, the expression vanishes identically for all sufficiently large t , uniformly in $u \in K$, and the case $z_i < 0$ is identical. Summing over the coordinates and using uniformity in u gives

$$\lim_{t \rightarrow \infty} F_z(t) = \sup_{u \in K} \langle \text{sgn}(z), u - x \rangle.$$

Since Z has independent standard Cauchy coordinates, $\text{sgn}(Z)$ is a Rademacher vector. Dominated convergence, with the dominating constant D_x , yields

$$\begin{aligned} \lim_{t \rightarrow \infty} \Phi_x(t) &= \mathbb{E} \sup_{u \in K} \langle \varepsilon, u - x \rangle \\ &= \mathbb{E} \sup_{u \in T} \langle \varepsilon, u \rangle - \mathbb{E} \langle \varepsilon, x \rangle \\ &= b(T). \end{aligned}$$

Here we used that a linear functional has the same supremum over T and over $\text{conv}(T)$, and that $\mathbb{E}\varepsilon = 0$.

As a finite concave function, Φ_x is absolutely continuous on every compact subinterval of $(0, \infty)$. Thus, for $0 < a < b < \infty$,

$$\Phi_x(b) - \Phi_x(a) = \int_a^b \Phi'_x(t) dt.$$

The derivative is nonnegative almost everywhere because Φ_x is nondecreasing. Letting $a \downarrow 0$ and $b \uparrow \infty$, and using the continuity at zero and the endpoint limit proved above, gives

$$\int_0^\infty \Phi'_x(t) dt = \lim_{b \rightarrow \infty} \Phi_x(b) - \Phi_x(0) = b(T).$$

□

Remark 3.4. The preceding proof applies whenever Z has i.i.d. coordinates with a symmetric continuous distribution. The Cauchy assumption becomes essential in the next subsection.

3.2. The potential's derivative dominates the error of a leave-one-coordinate-out estimator. We next connect the derivative of $\Phi_x(t)$ with the error of a leave-one-coordinate-out estimator in the Cauchy channel.

Fix $i \in [n]$ and $w \in \mathbb{R}^{n-1}$. Define

$$g_{i,w}(u) := d_K(w_1, \dots, w_{i-1}, u, w_i, \dots, w_{n-1}). \quad (25)$$

The function $g_{i,w}$ is convex, 1-Lipschitz, and coercive, so its set of minimizers is a nonempty compact interval. Denote its left endpoint by $\zeta_i(w)$. This is a Borel measurable tie-breaking rule: every minimizer lies in the fixed compact interval

$$I_i := \left[\min_{v \in K} v_i, \max_{v \in K} v_i \right],$$

and the measurable maximum theorem applied to the jointly continuous map $(w, u) \mapsto g_{i,w}(u)$ shows that the left endpoint of its argmin over I_i is measurable.

For $y = (y_1, \dots, y_n) \in \mathbb{R}^n$, write

$$y_{-i} := (y_1, \dots, y_{i-1}, y_{i+1}, \dots, y_n).$$

Define

$$\widehat{x}^{\text{loc}}(y) := \zeta_i(y_{-i}), \quad \widehat{x}^{\text{loc}}(y) := (\widehat{x}_1^{\text{loc}}(y), \dots, \widehat{x}_n^{\text{loc}}(y)). \quad (26)$$

Each coordinate ignores the corresponding coordinate of y , and the measurability of the functions ζ_i shows that $\widehat{x}^{\text{loc}} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a Borel measurable estimator. This coordinatewise estimator need not belong to K and should not be confused with the K -valued selector Π_K above.

Remark 3.5. We interpret $\widehat{x}_i^{\text{loc}}(y)$ as a leave-one-coordinate-out ℓ_1 -estimator of x_i from the Cauchy observation $Y = y = x + tZ$. It chooses the i -th coordinate substitution that minimizes the ℓ_1 -distance to K . This estimator arises naturally from the derivative of the potential $\Phi_x(t)$.

The following key lemma shows that, for almost every $t > 0$, the derivative of $\Phi_x(t)$ controls the expected capped quadratic error of this estimator, measured by φ_t from [Eq. \(8\)](#).

Lemma 3.6. *For any $x \in T$ and almost every $t > 0$,*

$$\Phi'_x(t) \geq \frac{1}{4\pi} \mathbb{E}_Z \varphi_t(x, \widehat{x}^{\text{loc}}(x + tZ)). \quad (27)$$

Proof. For fixed $z \in \mathbb{R}^n$, let

$$F_z(t) := t\|z\|_1 - d_K(x + tz).$$

The function F_z is nondecreasing and concave. It is also locally absolutely continuous, since

$$|F_z(t) - F_z(s)| \leq 2|t - s|\|z\|_1.$$

Let $q_z(t) := F'_z(t)$ at points where the derivative exists, and set $q_z(t) := 0$ on the remaining null set. Then $q_z(t) \geq 0$.

The convex function d_K is differentiable Lebesgue almost everywhere on $\mathbb{R}^n$. Moreover, because it is 1-Lipschitz with respect to $\|\cdot\|_1$, every gradient at a differentiability point

satisfies $\|\nabla d_K\|_\infty \leq 1$. For every $t > 0$, the random vector $x + tZ$ has an absolutely continuous distribution. Consequently, Fubini's theorem and the chain rule give, for $dt \otimes P_Z$ -almost every (t, z) ,

$$q_z(t) = \sum_{i=1}^n [|z_i| - z_i \partial_i d_K(x + tz)].$$

Every summand on the right is nonnegative.

For $0 < a < b < \infty$, absolute continuity of F_Z and Tonelli's theorem therefore give

$$\begin{aligned} \Phi_x(b) - \Phi_x(a) &= \mathbb{E}_Z [F_Z(b) - F_Z(a)] \\ &= \mathbb{E}_Z \int_a^b q_Z(t) dt \\ &= \int_a^b \mathbb{E}_Z q_Z(t) dt. \end{aligned}$$

It follows that, for almost every $t > 0$,

$$\Phi'_x(t) = \mathbb{E}_Z \sum_{i=1}^n [|Z_i| - Z_i \partial_i d_K(x + tZ)]. \quad (28)$$

This argument does not require $\mathbb{E}|Z_i| < \infty$.

Fix such a t and a coordinate i . Write

$$W := Y_{t,-i} = x_{-i} + tZ_{-i}.$$

By Fubini's theorem, for P_W -almost every w , the identity

$$\partial_i d_K(w, x_i + tZ_i) = g'_{i,w}(x_i + tZ_i)$$

holds almost surely in Z_i . Notice also that Z_i is independent of W and remains a standard Cauchy variable after conditioning on $W = w$.

Fix such a w , and write

$$\zeta := \zeta_i(w), \quad r := \frac{|x_i - \zeta|}{t}.$$

The function $g_{i,w}$ is convex and 1-Lipschitz, so $|g'_{i,w}| \leq 1$ wherever its derivative exists. In particular,

$$|z| - z g'_{i,w}(x_i + tz) \geq 0$$

for almost every $z \in \mathbb{R}$.

Suppose first that $x_i \leq \zeta$. For $0 \leq z \leq r$, one has $x_i + tz \leq \zeta$. Since ζ is the left endpoint of the minimizer interval of $g_{i,w}$, convexity gives

$$g'_{i,w}(x_i + tz) \leq 0$$

for almost every $z \in [0, r]$. Hence

$$|z| - z g'_{i,w}(x_i + tz) \geq z$$

on this interval, almost everywhere. Using the standard Cauchy density and discarding the nonnegative contribution from its complement, we obtain

$$\begin{aligned} \mathbb{E}_{Z_i} \left[|Z_i| - Z_i g'_{i,w}(x_i + tZ_i) \right] \\ \geq \frac{1}{\pi} \int_0^r \frac{z}{1+z^2} dz = \frac{1}{2\pi} \log(1+r^2). \end{aligned} \quad (29)$$

If $x_i \geq \zeta$, the same argument applies on $[-r, 0]$, where a derivative of $g_{i,w}$ is nonnegative.

Since $\hat{x}_i^{\text{loc}}(Y_t) = \zeta_i(W)$, we have shown that, for P_W -almost every w ,

$$\begin{aligned} \mathbb{E}_{Z_i} [|Z_i| - Z_i \partial_i d_K(x + tZ) | W = w] \\ \geq \frac{1}{2\pi} \log \left(1 + \frac{|x_i - \hat{x}_i^{\text{loc}}(Y_t)|^2}{t^2} \right). \end{aligned} \quad (30)$$

For every $r \geq 0$,

$$\frac{1}{2\pi} \log(1+r^2) \geq \frac{1}{4\pi} (1 \wedge r^2). \quad (31)$$

Indeed, $\log(1+r^2) \geq r^2/2$ when $r \leq 1$, while $\log(1+r^2) \geq \log 2 > 1/2$ when $r \geq 1$. Averaging over W , summing over i , and using [Eq. \(28\)](#) now yields

$$\Phi'_x(t) \geq \frac{1}{4\pi} \mathbb{E}_Z \varphi_t(x, \hat{x}^{\text{loc}}(x + tZ)).$$

□

3.3. The Bernoulli width upper bounds the cMMSE integral. The preceding bounds hold for every fixed planted point $x \in T$. Now let $X \sim \mu$, where $\mu \in \mathcal{P}(T)$, and regard μ as the prior in the Cauchy channel.

For completeness, the function $t \mapsto \text{cMMSE}_\mu(t)$ is Borel measurable on $(0, \infty)$. Indeed, let

$$J_T := \prod_{i=1}^n \left[\min_{x \in T} x_i, \max_{x \in T} x_i \right].$$

Coordinatewise projection of an action onto J_T cannot increase $\varphi_t(x, \cdot)$ for any $x \in T$, so the conditional Bayes-action optimization may be restricted to the compact set J_T . Since T is finite, the Cauchy posterior weights are Borel functions of (t, y) , and the conditional risk is continuous in the action. The measurable minimum theorem therefore shows that the conditional Bayes risk, and hence $\text{cMMSE}_\mu(t)$, is Borel measurable.

Optimizing over all estimators instead of using the leave-one-coordinate-out estimator in [Lemma 3.6](#) yields [Eq. \(16\)](#).

Proposition 3.7 (cMMSE-area inequality). *For every $\mu \in \mathcal{P}(T)$,*

$$b(T) \geq \frac{1}{4\pi} \int_0^\infty \text{cMMSE}_\mu(t) dt. \quad (32)$$

Proof. For each $x \in T$, [Lemma 3.6](#) holds outside a Lebesgue null set $N_x \subset (0, \infty)$. Since T is finite,

$$N := \bigcup_{x \in T} N_x$$

is again a null set. Thus, for every $t \notin N$, we may average the pivotal inequality simultaneously over $X \sim \mu$ to obtain

$$\mathbb{E}_X \Phi'_X(t) \geq \frac{1}{4\pi} \mathbb{E} \varphi_t(X, \hat{x}^{\text{loc}}(Y_t)) \geq \frac{1}{4\pi} \text{cMMSE}_\mu(t).$$

The second inequality holds because $\hat{x}^{\text{loc}}$ is a Borel measurable estimator and is therefore feasible in [Eq. \(12\)](#).

Finally, $\Phi'_x(t) \geq 0$ almost everywhere. Tonelli's theorem and [Lemma 3.3](#) give

$$\begin{aligned} \int_0^\infty \mathbb{E}_X \Phi'_X(t) dt &= \mathbb{E}_X \int_0^\infty \Phi'_X(t) dt \\ &= \mathbb{E}_X b(T) = b(T). \end{aligned}$$

Consequently,

$$b(T) \geq \frac{1}{4\pi} \int_0^\infty \text{cMMSE}_\mu(t) dt.$$

□

4. FROM THE CAUCHY cMMSE TO RATE-DISTORTION

The goal of this section is to prove (17), the second comparison in [Theorem 2.2](#). Together with [Section 3](#), this completes the proof of that theorem. The key input is a Cauchy information-estimation inequality, proved in [Appendix A](#), that plays the role of the I-MMSE identity for the Gaussian additive channel. A posterior-replica coupling then converts this information bound into the required rate-distortion estimate.

For $t \geq 0$, consider the Cauchy observation $Y_t = X + tZ$, where $X \sim \mu$ is independent of Z , and let

$$I_\mu(t) := I(X; Y_t). \tag{33}$$

Proposition 4.1. *For every $\mu \in \mathcal{P}(T)$ and $t > 0$,*

$$I_\mu(t) \leq 16 \int_t^\infty \frac{\text{cMMSE}_\mu(s)}{s} ds. \tag{34}$$

This is the second assertion of [Proposition A.2](#), proved in [Appendix A](#) by a direct calculation using the harmonicity of the Cauchy density.

For $t > 0$, let $X_t^{(1)}, X_t^{(2)}$ be conditionally independent vector-valued draws from the posterior law $P_{X|Y_t}$. Define

$$\begin{aligned} \text{rcMMSE}_\mu(t) &:= \mathbb{E} \varphi_t(X_t^{(1)}, X_t^{(2)}) = \sum_{i=1}^n \text{rcMMSE}_{i,\mu}(t), \\ \text{rcMMSE}_{i,\mu}(t) &:= \mathbb{E} \left[1 \wedge \frac{|X_{t,i}^{(1)} - X_{t,i}^{(2)}|^2}{t^2} \right]. \end{aligned} \quad (35)$$

The following lemma compares this posterior-replica risk pointwise with the capped Bayes risk.

Lemma 4.2. *For every $t > 0$, $\text{rcMMSE}_\mu(t) \leq 4\text{cMMSE}_\mu(t)$.*

Proof. Notice that for any real numbers x, x', a we have

$$1 \wedge \frac{|x - x'|^2}{t^2} \leq 2 \left(1 \wedge \frac{|x - a|^2}{t^2} \right) + 2 \left(1 \wedge \frac{|x' - a|^2}{t^2} \right). \quad (36)$$

Indeed, $|x - x'|^2 \leq 2|x - a|^2 + 2|x' - a|^2$, and $1 \wedge (2u + 2v) \leq 2(1 \wedge u) + 2(1 \wedge v)$. Summing Eq. (36) over coordinates gives

$$\varphi_t(x, x') \leq 2\varphi_t(x, a) + 2\varphi_t(x', a).$$

Hence, for every Y_t -measurable function $a(Y_t)$, the Nishimori identity gives

$$\mathbb{E}[\varphi_t(X_t^{(1)}, X_t^{(2)}) | Y_t] = \mathbb{E}[\varphi_t(X_t^{(1)}, X) | Y_t] \leq 4\mathbb{E}[\varphi_t(X, a(Y_t)) | Y_t].$$

Averaging over Y_t and optimizing over a concludes the proof. $\square$

Equipped with Proposition 4.1 and Lemma 4.2 we are now in a position to prove Eq. (17).

Proposition 4.3 (Cauchy Bayes-risk area controls rate distortion). *For every $\mu \in \mathcal{P}(T)$,*

$$\int_0^\infty \text{RD}_\mu(t) dt \leq 20 \int_0^\infty \text{cMMSE}_\mu(t) dt. \quad (37)$$

Proof. Fix $t > 0$ and consider the Cauchy observation $Y_t = X + tZ$ with prior $X \sim \mu$. Let $X_t^{(1)}$ be a draw from $P_{X|Y_t}$ sampled conditionally independently of X given Y_t . By Bayes' rule, $X_t^{(1)}$ has marginal law μ , and therefore $(X, X_t^{(1)})$ is a valid coupling. Hence

$$I(X; X_t^{(1)}) \leq I(X; Y_t) = I_\mu(t), \quad (38)$$

where the inequality follows from the Markov chain $X \rightarrow Y_t \rightarrow X_t^{(1)}$. Also, by the Nishimori identity,

$$\mathbb{E} \varphi_t(X, X_t^{(1)}) = \text{rcMMSE}_\mu(t), \quad (39)$$

Therefore, combining this with Proposition 4.1,

$$\text{RD}_\mu(t) \leq I_\mu(t) + \text{rcMMSE}_\mu(t) \leq \text{rcMMSE}_\mu(t) + 16 \int_t^\infty \frac{\text{cMMSE}_\mu(s)}{s} ds. \quad (40)$$

Integrating [Eq. \(40\)](#), applying Tonelli's theorem, and using [Lemma 4.2](#),

$$\begin{aligned} \int_0^\infty \text{RD}_\mu(t) dt &\leq \int_0^\infty \text{rcMMSE}_\mu(t) dt + 16 \int_0^\infty \int_t^\infty \frac{\text{cMMSE}_\mu(s)}{s} ds dt \\ &\leq 20 \int_0^\infty \text{cMMSE}_\mu(s) ds. \end{aligned}$$

This proves [Eq. \(37\)](#). $\square$

Proof of [Theorem 2.2](#). The first inequality is [Proposition 3.7](#), and the second is [Proposition 4.3](#). Combining them gives

$$\int_0^\infty \text{RD}_\mu(t) dt \leq 20 \int_0^\infty \text{cMMSE}_\mu(t) dt \leq 80\pi b(T).$$

$\square$

5. FROM THE RATE-DISTORTION INTEGRAL TO DISTRIBUTIONAL DECOMPOSITION

The goal of this section is to prove the prescribed-law estimate $\delta_T(\mu) \lesssim \int_0^\infty \text{RD}_\mu(t) dt$ and then use a minimax argument to derive $\Lambda(T) \lesssim R(T)$, which is [Theorem 2.3](#). We state the two required lemmas, deduce the theorem, and then prove the lemmas. The prescribed-law estimate will also supply the final comparison in [Section 6](#).

5.1. Two key lemmas. We use the Gaussian width W from [\(1\)](#). The distributional functionals G and δ_T were defined previously.

The first lemma bounds $\delta_T(\mu)$ by the rate-distortion integral for every $\mu \in \mathcal{P}(T)$.

Lemma 5.1. *For every $\mu \in \mathcal{P}(T)$,*

$$\delta_T(\mu) \leq 30 \int_0^\infty \text{RD}_\mu(t) dt. \quad (41)$$

The second, a minimax lemma, places $\sup_{\mu \in \mathcal{P}(T)} \delta_T(\mu)$ in a form directly comparable with the Bernoulli functional $\Lambda(T)$.

Lemma 5.2. *For every finite $T \subset \mathbb{R}^n$,*

$$\inf_{a:T \rightarrow \mathbb{R}^n} \left\{ \max_{x \in T} \|a(x)\|_1 + W(\{x - a(x) : x \in T\}) \right\} \leq 2 \sup_{\mu \in \mathcal{P}(T)} \delta_T(\mu).$$

5.2. Proof of the rate-distortion lower bound.

Proof of [Theorem 2.3](#). By [Lemmas 5.1](#) and [5.2](#),

$$\inf_{a:T \rightarrow \mathbb{R}^n} \left\{ \max_{x \in T} \|a(x)\|_1 + W(\{x - a(x) : x \in T\}) \right\} \leq 60 \sup_{\mu \in \mathcal{P}(T)} \int_0^\infty \text{RD}_\mu(t) dt = 60R(T).$$

Fix $\varepsilon > 0$, and choose $a : T \rightarrow \mathbb{R}^n$ such that, with

$$A := a(T), \quad U := \{x - a(x) : x \in T\}, \quad M := \max_{x \in T} \|a(x)\|_1,$$

we have $M + W(U) \leq 60R(T) + \varepsilon$. Since $T \subseteq A + U$ and $\sup_{z \in A} \|z\|_1 = M$, the Gaussian majorizing-measure theorem [Tal87] gives a universal constant $C_G > 0$ such that

$$\gamma_2(U, \|\cdot\|_2) \leq C_G W(U).$$

Consequently, (5) yields, with $C_0 := \max\{1, C_G\}$,

$$\Lambda(T) \leq M + \gamma_2(U, \|\cdot\|_2) \leq M + C_G W(U) \leq C_0(60R(T) + \varepsilon).$$

Letting $\varepsilon \downarrow 0$ proves $R(T) \geq (60C_0)^{-1}\Lambda(T)$, and hence (19). $\square$

5.3. The distributional decomposition lemma. We now prove Lemma 5.1, which supplies the last comparison in the prescribed-law cycle. We first prove a discrete multiscale bound and then compare the resulting dyadic sum with the rate-distortion integral. For $k \in \mathbb{Z}$, set

$$r_k := 2^{-k}.$$

The following construction is based on the same general decomposition-from-chaining principle as [Tal94, Proposition 4.3] and [BL14, Theorem 3.1]: a multiscale, chaining-type description is converted into an ℓ_1 -plus-Gaussian decomposition. Here the multiscale data are supplied by rate-distortion self-couplings.

Lemma 5.3. *Let $\mu \in \mathcal{P}(T)$, let $X \sim \mu$, and let $k_0 < k_1$ be integers such that*

$$r_{k_0} \geq 2 \operatorname{diam}_\infty(T).$$

Suppose that $U_{k_0}, \dots, U_{k_1}$ are conditionally independent given X and satisfy $U_k \sim \mu$ for every $k = k_0, \dots, k_1$. Assume in addition that U_{k_0} is independent of X and that $U_{k_1} = X$. For $k = k_0, \dots, k_1$, set

$$d_k := \mathbb{E}\varphi_{r_k}(X, U_k), \quad q_k := I(X; U_k).$$

Then

$$\delta_T(\mu) \leq \frac{17}{2} \sum_{k=k_0+1}^{k_1-1} r_k d_k + \frac{9}{2\sqrt{2}} \sum_{k=k_0}^{k_1} r_k (d_k + q_k). \quad (42)$$

Proof. Let clip_R act coordinatewise by truncation to $[-R, R]$, and define for $k = k_0, \dots, k_1 - 1$,

$$\Delta_k := \operatorname{clip}_{3r_k/2}(U_{k+1} - U_k), \quad \widetilde{V} := U_{k_0} + \sum_{k=k_0}^{k_1-1} \Delta_k.$$

For $x \in \operatorname{supp} \mu$, define

$$V(x) := \mathbb{E}[\widetilde{V} \mid X = x], \quad A(x) := x - V(x).$$

On any μ -null point of T , set $A(x) = 0$ and $V(x) = x$. Thus $x = A(x) + V(x)$ on T .

We first bound A . Fix a coordinate $j = 1, \dots, n$, and define for $k = k_0, \dots, k_1$, $v_k := U_{k,j} - X_j$. We call k bad when $|v_k| > r_k$, and good otherwise.

Notice that the endpoint indices k_0, k_1 are good by definition. Now, suppose that $p, \dots, q$ is any maximal run of bad indices; that is, $p, \dots, q$ are bad and $p-1$ and $q+1$ are good. Define for $k = k_0, \dots, k_1 - 1$ and $j = 1, \dots, n$,

$$E_{k,j} := (U_{k+1,j} - U_{k,j}) - \Delta_{k,j}. \quad (43)$$

Now, for any feasible value of k , if k and $k+1$ are good, then

$$|U_{k+1,j} - U_{k,j}| \leq r_k + r_{k+1} = \frac{3}{2}r_k,$$

so $E_{k,j} = 0$. Then from (43) we obtain

$$\left| \sum_{k=p-1}^q E_{k,j} \right| \leq r_{p-1} + r_{q+1} + \frac{3}{2} \sum_{k=p-1}^q r_k \leq \frac{17}{2}r_p. \quad (44)$$

Since $X_j - \tilde{V}_j = \sum_{k=k_0}^{k_1-1} E_{k,j}$, we may decompose $\{k_0+1, \dots, k_1-1\}$ into disjoint blocks of consecutive bad indices (maximal bad runs), and use (44) to obtain that for any $j = 1, \dots, n$,

$$|X_j - \tilde{V}_j| \leq \frac{17}{2} \sum_{k=k_0+1}^{k_1-1} r_k \mathbf{1}_{\{|X_j - U_{k,j}| > r_k\}}$$

and hence

$$\|X - \tilde{V}\|_1 \leq \frac{17}{2} \sum_{k=k_0+1}^{k_1-1} r_k \sum_{j=1}^n \mathbf{1}_{\{|X_j - U_{k,j}| > r_k\}} \leq \frac{17}{2} \sum_{k=k_0+1}^{k_1-1} r_k \varphi_{r_k}(X, U_k). \quad (45)$$

The last inequality holds because every indicated coordinate contributes 1 to $\varphi_{r_k}(X, U_k)$. Applying Jensen's inequality to (45) gives

$$\mathbb{E} \|A(X)\|_1 = \mathbb{E} \|X - \mathbb{E}[\tilde{V}|X]\|_1 \leq \frac{17}{2} \sum_{k=k_0+1}^{k_1-1} r_k d_k. \quad (46)$$

For the Gaussian part, the coordinatewise definition of clipping gives

$$\|\Delta_k\|_2^2 \leq \frac{9}{4} r_k^2 (\varphi_{r_k}(X, U_k) + \varphi_{r_{k+1}}(X, U_{k+1})). \quad (47)$$

To verify this coordinatewise, put

$$\alpha := \frac{|X_j - U_{k,j}|}{r_k}, \quad \beta := \frac{|X_j - U_{k+1,j}|}{r_{k+1}}.$$

If $\alpha > 1$ or $\beta > 1$, the corresponding coordinate on the right-hand side of (47) is at least $\frac{9}{4}r_k^2$, which bounds $|\Delta_{k,j}|^2$ by clipping. If $\alpha, \beta \leq 1$, then

$$|\Delta_{k,j}|^2 \leq |U_{k+1,j} - U_{k,j}|^2 \leq r_k^2 \left(\alpha + \frac{\beta}{2} \right)^2 \leq \frac{9}{4} r_k^2 (\alpha^2 + \beta^2).$$

This proves (47).

Take now an arbitrary coupling of $X \sim \mu$ with a standard Gaussian G_0 , as in the pushforward representation of $G(V_{\#}\mu)$. Conditionally on X , sample the U_k 's according to their original product kernel, independently of G_0 . Then $\mathbb{E}[\tilde{V} | X, G_0] = V(X)$, so

$$\mathbb{E} \langle V(X), G_0 \rangle = \mathbb{E} \langle \tilde{V}, G_0 \rangle = \sum_{k=k_0}^{k_1-1} \mathbb{E} \langle \Delta_k, G_0 \rangle;$$

the U_{k_0} -term vanishes because U_{k_0} is independent of G_0 . Moreover, data processing and conditional independence give

$$I(G_0; U_k, U_{k+1}) \leq I(X; U_k, U_{k+1}) \leq q_k + q_{k+1}.$$

The Gaussian mean–information inequality gives

$$\mathbb{E} \|\mathbb{E}[G_0 \mid U_k, U_{k+1}]\|_2^2 \leq 2I(G_0; U_k, U_{k+1});$$

indeed, this follows by applying the following inequality

$$D_{\text{KL}}(Q \| N(0, I_n)) \geq \frac{1}{2} \|\mathbb{E}_Q G_0\|_2^2, \quad \forall Q$$

with $Q = P_{G_0|U_k, U_{k+1}}$, and then taking the expectation with respect to $P_{U_k, U_{k+1}}$. Since Δ_k is a function of (U_k, U_{k+1}) , Cauchy–Schwarz and (47) yield

$$\begin{aligned} \mathbb{E} \langle \Delta_k, G_0 \rangle &= \mathbb{E} \langle \Delta_k, \mathbb{E}[G_0 \mid U_k, U_{k+1}] \rangle \\ &\leq (\mathbb{E} \|\Delta_k\|_2^2)^{1/2} (\mathbb{E} \|\mathbb{E}[G_0 \mid U_k, U_{k+1}]\|_2^2)^{1/2} \\ &\leq \frac{3}{\sqrt{2}} r_k \sqrt{(d_k + d_{k+1})(q_k + q_{k+1})} \\ &\leq \frac{3}{2\sqrt{2}} r_k (d_k + d_{k+1} + q_k + q_{k+1}). \end{aligned}$$

Writing $s_k := d_k + q_k$, using $r_{k-1} = 2r_k$, summing in k , and then taking the supremum over couplings of X and G_0 , we obtain

$$\begin{aligned} G(V_{\#}\mu) &\leq \frac{3}{2\sqrt{2}} \sum_{k=k_0}^{k_1-1} r_k (s_k + s_{k+1}) \\ &\leq \frac{9}{2\sqrt{2}} \sum_{k=k_0}^{k_1} r_k s_k. \end{aligned} \tag{48}$$

Combining (46) and (48) proves (42). $\square$

We are now in a position to prove [Lemma 5.1](#).

Proof of Lemma 5.1. The function $t \mapsto \text{RD}_{\mu}(t)$ is clearly nonincreasing. Integrating over each interval $[r_{k+1}, r_k]$, $k \in \mathbb{Z}$ therefore gives

$$\sum_{k \in \mathbb{Z}} r_k \text{RD}_{\mu}(r_k) \leq 2 \int_0^{\infty} \text{RD}_{\mu}(t) dt. \tag{49}$$

Removing μ -null points from T changes neither side of (41), so assume $\text{supp } \mu = T$. Fix integers $k_0 < k_1$, with k_0 sufficiently negative that $r_{k_0} \geq 2 \text{diam}_{\infty}(T)$. The finite self-coupling polytope is compact and the objective in (9) is continuous, so for every $k_0 < k < k_1$ we may choose a self-coupling (X, U_k) attaining $\text{RD}_{\mu}(r_k)$. Combine the corresponding kernels conditionally independently given X , and complete the family with an independent copy U_{k_0}

and with $U_{k_1} = X$. For $k_0 \leq k \leq k_1$, let d_k and q_k be the quantities associated with this family in [Lemma 5.3](#). At every interior index,

$$d_k + q_k = \text{RD}_\mu(r_k),$$

whereas the endpoint quantities satisfy

$$q_{k_0} = 0, \quad d_{k_1} = 0, \quad q_{k_1} = H(\mu).$$

Since $d_k \leq d_k + q_k$ and $\frac{17}{2} + \frac{9}{2\sqrt{2}} < 15$, [Lemma 5.3](#) gives

$$\delta_T(\mu) \leq 15 \left[\sum_{k=k_0+1}^{k_1-1} r_k \text{RD}_\mu(r_k) + r_{k_0} d_{k_0} + r_{k_1} H(\mu) \right].$$

If X' is an independent copy of X , then since T is bounded,

$$r_{k_0} d_{k_0} \leq \frac{\mathbb{E} \|X - X'\|_2^2}{r_{k_0}} \longrightarrow 0 \quad (k_0 \rightarrow -\infty),$$

while $r_{k_1} H(\mu) \rightarrow 0$ as $k_1 \rightarrow +\infty$. Letting the endpoints tend to $-\infty$ and $+\infty$, respectively, and using [\(49\)](#) proves the claim. $\square$

5.4. The minimax decomposition lemma. We next prove [Lemma 5.2](#). This step converts the prior-wise decompositions furnished by [Lemma 5.1](#) into the single set-level decomposition required for [Theorem 2.3](#).

Proof of Lemma 5.2. For $a : T \rightarrow \mathbb{R}^n$, put $v_a := \text{id}_T - a$ and define

$$J(a, \mu) := \sum_{x \in T} \mu(x) \|a(x)\|_1 + G((v_a)_\# \mu).$$

Then $\delta_T(\mu) = \inf_a J(a, \mu)$. By the pushforward representation of G , for fixed μ ,

$$G((v_a)_\# \mu) = \sup_{\substack{X \sim \mu, G_0 \sim N(0, I_n) \\ \text{jointly coupled}}} \mathbb{E} \langle v_a(X), G_0 \rangle.$$

The right-hand side is a supremum of continuous affine functions of a and is finite everywhere, so it is convex and continuous on the finite-dimensional space $(\mathbb{R}^n)^T$. Hence $a \mapsto J(a, \mu)$ is convex and lower semicontinuous.

For fixed a , the map $\mu \mapsto J(a, \mu)$ is concave because mixtures of feasible couplings are feasible. It is upper semicontinuous as well. Indeed, if $\mu_k \rightarrow \mu$, near-optimal couplings along a subsequence realizing the limsup are tight because their Gaussian marginal is fixed. Any weak limit is feasible for μ , while $|\langle v_a(X), G_0 \rangle| \leq M_a \|G_0\|_2$, where $M_a := \max_{x \in T} \|v_a(x)\|_2$, gives uniform integrability. Passing to the limit in the coupling objective and using the continuity of the first term in $J(a, \mu)$ proves upper semicontinuity.

The set $\mathcal{P}(T)$ is compact and convex, while $(\mathbb{R}^n)^T$ is convex. Sion's minimax theorem therefore yields

$$\inf_a \sup_{\mu \in \mathcal{P}(T)} J(a, \mu) = \sup_{\mu \in \mathcal{P}(T)} \inf_a J(a, \mu) = \sup_{\mu \in \mathcal{P}(T)} \delta_T(\mu). \quad (50)$$

Fix a . Taking μ to be a point mass at a maximizer of $\|a(x)\|_1$ gives

$$\sup_{\mu \in \mathcal{P}(T)} J(a, \mu) \geq \max_{x \in T} \|a(x)\|_1.$$

Let $x(G_0)$ be a measurable maximizer of $\langle v_a(x), G_0 \rangle$, and let μ_a be its law. The coupling $(v_a(x(G_0)), G_0)$ then gives

$$G((v_a)_{\#}\mu_a) \geq \mathbb{E} \max_{x \in T} \langle v_a(x), G_0 \rangle = W(v_a(T)).$$

It follows that

$$\max_{x \in T} \|a(x)\|_1 + W(v_a(T)) \leq 2 \sup_{\mu \in \mathcal{P}(T)} J(a, \mu).$$

Taking the infimum over a and using (50) proves the lemma. $\square$

6. PROOF OF THE DISTRIBUTIONAL THEOREM

The goal of this section is to complete [Theorem 2.1](#). Type lifting, meaning passage from a law to the uniform law on sequences with that empirical distribution, combines with [Theorem 2.2](#) to give $\int_0^\infty \text{RD}_\mu(t) dt \lesssim B(\mu)$, elementary Gaussianization gives $B(\mu) \lesssim \delta_T(\mu)$, and [Lemma 5.1](#) supplies the remaining comparison $\delta_T(\mu) \lesssim \int_0^\infty \text{RD}_\mu(t) dt$. Taking the supremum over μ and invoking [Theorem 2.3](#) then completes the set-level assertion.

Lemma 6.1 (Lifting rate–distortion inequality). *For every finite $T \subset \mathbb{R}^n$ and every $\mu \in \mathcal{P}(T)$,*

$$B(\mu) \gtrsim \int_0^\infty \text{RD}_\mu(t) dt.$$

Proof. Call $\mu \in \mathcal{P}(T)$ rational if $\mu(x) \in \mathbb{Q}$ for every $x \in T$. We first treat rational μ . Call a positive integer N compatible with μ if $N\mu(x)$ is an integer for every x . For a sequence $\mathbf{x} = (x_1, \dots, x_N) \in T^N$, its empirical measure

$$L_N(\mathbf{x}) := \frac{1}{N} \sum_{j=1}^N \delta_{x_j}$$

is called its *type* in information theory. The corresponding *exact type class* is the fiber

$$\mathcal{T}_N(\mu) := \{\mathbf{x} \in T^N : L_N(\mathbf{x}) = \mu\}.$$

Thus $\mathcal{T}_N(\mu)$ consists of the sequences containing exactly $N\mu(x)$ copies of each x . Let μ_N be its uniform law, equivalently, the conditional law of an i.i.d. μ -sample given $L_N = \mu$.

Applied to the Bernoulli process, the type lifting identity [[Liu25](#), Lemma 5]; see also [[vH25](#), Proposition 3.1], gives

$$B(\mu) = \lim_{\substack{N \rightarrow \infty \\ N \text{ compatible with } \mu}} \frac{1}{N} b(\mathcal{T}_N(\mu)). \quad (51)$$

We next quantify the small entropy loss caused by conditioning on the type. Set

$$m := |\text{supp}(\mu)|, \quad r_N := NH(\mu) - \log |\mathcal{T}_N(\mu)|.$$

If K is the count vector of an i.i.d. μ -sample and $n_x := N\mu(x)$ for $x \in \text{supp}(\mu)$, then

$$\mathbb{P}(K = k) = \frac{N!}{N^N} \prod_{x \in \text{supp}(\mu)} \frac{n_x^{k_x}}{k_x!}.$$

Since $k \mapsto n^k/k!$ is maximized at $k = n$ (and at $n - 1$), $K = (n_x)_x$ is a mode. There are at most $(N + 1)^{m-1}$ possible count vectors, while

$$\mathbb{P}\{L_N = \mu\} = |\mathcal{T}_N(\mu)|e^{-NH(\mu)} = e^{-r_N}.$$

Consequently,

$$0 \leq r_N \leq (m - 1) \log(N + 1). \quad (52)$$

Regard $\mathcal{T}_N(\mu)$ as a subset of $(\mathbb{R}^n)^N \cong \mathbb{R}^{nN}$ and interpret $\text{RD}_{\mu_N}(t)$ according to (9) in this ambient space. In particular,

$$\varphi_t(\mathbf{x}, \mathbf{u}) = \sum_{j=1}^N \varphi_t(x_j, u_j).$$

Let $(\mathbf{X}, \mathbf{U})$ be any self-coupling of μ_N , and let J be independent and uniform on $\{1, \dots, N\}$. Then $X_J, U_J \sim \mu$, and conditional-entropy subadditivity gives

$$\begin{aligned} I(\mathbf{X}; \mathbf{U}) &= NH(\mu) - r_N - H(\mathbf{X} | \mathbf{U}) \\ &\geq NH(\mu) - r_N - \sum_{j=1}^N H(X_j | U_j) \\ &= NH(\mu) - r_N - NH(X_J | U_J, J) \\ &\geq NI(X_J; U_J) - r_N. \end{aligned}$$

Moreover, $\mathbb{E}\varphi_t(\mathbf{X}, \mathbf{U}) = N\mathbb{E}\varphi_t(X_J, U_J)$. Taking the infimum over the self-coupling gives

$$\text{RD}_{\mu_N}(t) \geq N\text{RD}_{\mu}(t) - r_N. \quad (53)$$

Put $D := \text{diam}_2(T)$ and $s_N := \sqrt{N/(r_N + 1)}$. The independent self-coupling gives $\text{RD}_{\mu}(t) \leq D^2/t^2$. We therefore integrate (53) only up to the finite cutoff s_N , use the preceding bound for the tail, and apply Theorem 2.2:

$$\begin{aligned} \int_0^\infty \text{RD}_{\mu}(t) \, dt &\leq \frac{1}{N} \int_0^\infty \text{RD}_{\mu_N}(t) \, dt + \frac{s_N r_N}{N} + \frac{D^2}{s_N} \\ &\leq \frac{80\pi}{N} b(\mathcal{T}_N(\mu)) + (1 + D^2) \sqrt{\frac{r_N + 1}{N}}. \end{aligned}$$

Letting $N \rightarrow \infty$ through compatible integers and using (51) and (52) proves the lemma for rational μ .

For general μ , approximate it by rational laws. If $d := \frac{1}{2} \sum_{x \in T} |\nu(x) - \nu'(x)|$, a maximal-coupling kernel gives the first of the following bounds, and data processing gives the second:

$$|B(\nu) - B(\nu')| \leq \text{diam}_1(T)d, \quad |\text{RD}_{\nu}(t) - \text{RD}_{\nu'}(t)| \leq 2d \min \left\{ n, \frac{\text{diam}_2(T)^2}{t^2} \right\}.$$

Indeed, apply the kernel to the T -valued variable in the first coupling problem and independently to both variables in the second; each variable changes with probability d . The bound on the right is integrable over $(0, \infty)$, so both $B(\nu)$ and $\int_0^\infty \text{RD}_\nu(t) dt$ are continuous in ν . The rational case now completes the proof. $\square$

Proof of Theorem 2.1. By Lemma 6.1,

$$\int_0^\infty \text{RD}_\mu(t) dt \lesssim B(\mu).$$

For the reverse comparison, let ν be a finitely supported probability measure on $\mathbb{R}^n$ and fix a coupling of $Y \sim \nu$ and $\varepsilon \sim \text{Rad}^{\otimes n}$. If $g_1, \dots, g_n$ are independent standard Gaussian variables, independent of (Y, ε) , then $\tilde{G} := (\varepsilon_i |g_i|)_{i=1}^n \sim N(0, I_n)$ and

$$\mathbb{E}\langle Y, \tilde{G} \rangle = \sqrt{\frac{2}{\pi}} \mathbb{E}\langle Y, \varepsilon \rangle.$$

Taking the supremum over couplings gives $B(\nu) \leq \sqrt{\pi/2} G(\nu)$.

Now fix $a : T \rightarrow \mathbb{R}^n$ and put $\nu = (\text{id}_T - a)_\# \mu$. Since $\|\varepsilon\|_\infty = 1$, the preceding comparison gives

$$\begin{aligned} B(\mu) &\leq \mathbb{E}\|a(X)\|_1 + B(\nu) \\ &\leq \sqrt{\frac{\pi}{2}} \{\mathbb{E}\|a(X)\|_1 + G(\nu)\}. \end{aligned}$$

Taking the infimum over a yields $B(\mu) \leq \sqrt{\pi/2} \delta_T(\mu)$. Combining this with Lemma 5.1 gives

$$\int_0^\infty \text{RD}_\mu(t) dt \lesssim B(\mu) \lesssim \delta_T(\mu) \lesssim \int_0^\infty \text{RD}_\mu(t) dt,$$

so the prescribed-law assertion follows. Taking the supremum over $\mu \in \mathcal{P}(T)$, using (10) and the identity $\sup_{\mu \in \mathcal{P}(T)} B(\mu) = b(T)$ established above, gives $b(T) \asymp R(T)$. The elementary bound $b(T) \lesssim \Lambda(T)$, together with Theorem 2.3, therefore yields $b(T) \asymp R(T) \asymp \Lambda(T)$. This proves the set-level assertion. $\square$

APPENDIX A. A CAUCHY INFORMATION–ESTIMATION INEQUALITY

We prove the Cauchy information–estimation inequality, Proposition 4.1, used in Section 4. It is the analytic input for (17), the second comparison in Theorem 2.2.

A.1. A scalar Cauchy information–estimation inequality. First, notice that for any $x \in \mathbb{R}$ the PDF of $x + tZ$, where Z follows a Cauchy distribution is the Poisson kernel

$$c_t(y - x) = \frac{1}{\pi} \frac{t}{t^2 + (y - x)^2}, y \in \mathbb{R}. \quad (54)$$

It is easy to check that for any $x \in \mathbb{R}$, $c_t(y - x)$ as a function of $(y, t) \in \mathbb{R} \times (0, \infty)$ is harmonic, i.e., it satisfies for all $y, x \in \mathbb{R}$ and $t > 0$,

$$(\partial_y^2 + \partial_t^2)c_t(y - x) = 0. \quad (55)$$

The following one-dimensional information–estimation identity and consequence hold.

Lemma A.1 (Scalar Cauchy information-estimation identity). *Let U be a finitely supported real random variable, let S be an arbitrarily distributed random variable, and let Z be standard Cauchy and independent of (U, S) . Define*

$$F(t) := I(U; U + tZ \mid S). \quad (56)$$

Then F is decreasing and convex on $(0, \infty)$, with $\lim_{t \rightarrow \infty} F(t) = 0$. Moreover, let $U_t^{(1)}, U_t^{(2)}$ be i.i.d. draws from the posterior of U given $(S, U + tZ)$. Then

$$F''(t) \leq \frac{2}{t^2} \mathbb{E} \left[1 \wedge \frac{|U_t^{(1)} - U_t^{(2)}|^2}{t^2} \right]. \quad (57)$$

Consequently,

$$-F'(t) \leq 2 \int_t^\infty \frac{1}{s^2} \mathbb{E} \left[1 \wedge \frac{|U_s^{(1)} - U_s^{(2)}|^2}{s^2} \right] ds. \quad (58)$$

Proof. We first prove the result when S is deterministic. Fix a finite prior μ for U , and define

$$u_x(t, y) := c_t(y - x), \quad p(t, y) := \int_{-\infty}^{+\infty} u_x(t, y) \mu(dx).$$

Both u_x and p are positive harmonic functions of (y, t) . Also $I(U; U + tZ)$ equals

$$F(t) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} u_x(t, y) \log \frac{u_x(t, y)}{p(t, y)} dy \mu(dx) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \Psi(u_x(t, y), p(t, y)) dy \mu(dx) \quad (59)$$

where we defined, for positive variables u, v , $\Psi(u, v) := u \log(u/v)$. Direct calculations then give,

$$\Psi_{uu} = \frac{1}{u}, \quad \Psi_{uv} = -\frac{1}{v}, \quad \Psi_{vv} = \frac{u}{v^2}.$$

Let ∇ denote the two-dimensional gradient in (y, t) . Setting $u = u_x(t, y)$ and $v = p(t, y)$, which are harmonic in (y, t) , gives

$$\begin{aligned} (\partial_y^2 + \partial_t^2) \Psi(u, v) &= \frac{|\nabla u|^2}{u} - 2 \frac{\nabla u \cdot \nabla v}{v} + u \frac{|\nabla v|^2}{v^2} \\ &= u |\nabla \log u - \nabla \log v|^2, \end{aligned} \quad (60)$$

Let $J = [a, b] \Subset (0, \infty)$ and write $m_1 := \int x \mu(dx)$. Since μ has finite support, uniformly for $t \in J$ and $x \in \text{supp } \mu$, as $|y| \rightarrow \infty$,

$$\begin{aligned} u_x(t, y) &= \frac{t}{\pi y^2} \left(1 + \frac{2x}{y} + O_J(|y|^{-2}) \right), \\ p(t, y) &= \frac{t}{\pi y^2} \left(1 + \frac{2m_1}{y} + O_J(|y|^{-2}) \right). \end{aligned}$$

Hence

$$\log \frac{u_x(t, y)}{p(t, y)} = \frac{2(x - m_1)}{y} + O_J(|y|^{-2}).$$

Differentiating the finite rational expressions in t and y gives

$$|\partial_t^j \Psi(u_x(t, y), p(t, y))| \leq C_J(1 + |y|)^{-3}, \quad j = 0, 1, 2,$$

and

$$|\partial_y \Psi(u_x(t, y), p(t, y))| \leq C_J(1 + |y|)^{-4}.$$

On bounded y -intervals these bounds follow from smoothness and the strict positivity of p , uniformly for $t \in J$. Thus differentiation under the integral sign is justified by dominated convergence. Also, for every $R > 0$,

$$\int_{-R}^R \partial_y^2 \Psi(u_x, p) dy = \partial_y \Psi(u_x, p)(R) - \partial_y \Psi(u_x, p)(-R),$$

which tends to zero as $R \rightarrow \infty$ by the last bound. Therefore, after integrating (60) in y , the left-hand side is precisely $F''(t)$. Writing $Y_t := U + tZ$, we obtain

$$F''(t) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} u_x(t, y) |\nabla \log u_x(t, y) - \nabla \log p(t, y)|^2 dy \mu(dx),$$

or, equivalently,

$$F''(t) = \mathbb{E} |\nabla \log c_t(Y_t - U) - \nabla \log p(t, Y_t)|^2, \quad (61)$$

A direct differentiation implies

$$\nabla \log p(t, Y_t) = \mathbb{E} [\nabla \log c_t(Y_t - U) \mid Y_t]. \quad (62)$$

In particular, Eq. (61) is a conditional variance, which proves that F is convex. If $U_t^{(1)}, U_t^{(2)}$ are two posterior replicas of U given Y_t , then

$$F''(t) = \frac{1}{2} \mathbb{E} \left| \nabla \log c_t(Y_t - U_t^{(1)}) - \nabla \log c_t(Y_t - U_t^{(2)}) \right|^2. \quad (63)$$

We now compute the integrand. Setting $r = y - x$ and $v = r/t$, direct differentiation of the Poisson kernel Eq. (54) gives

$$t \nabla_{(y,t)} \log c_t(r) = V(v) := \left(-\frac{2v}{1+v^2}, \frac{v^2-1}{1+v^2} \right). \quad (64)$$

Note that the vector $V(v)$ lies on the unit circle. A direct calculation then gives

$$\|V(v) - V(w)\|^2 = \frac{4(v-w)^2}{(1+v^2)(1+w^2)} \leq 4(1 \wedge |v-w|^2). \quad (65)$$

Apply Eq. (64) with

$$v := \frac{Y_t - U_t^{(1)}}{t}, \quad w := \frac{Y_t - U_t^{(2)}}{t}.$$

Since

$$v - w = \frac{U_t^{(2)} - U_t^{(1)}}{t},$$

substituting Eq. (65) into Eq. (63) proves Eq. (57).

For $0 < t_1 < t_2$, let Z_1, Z_2 be independent standard Cauchy random variables, independent of U . Cauchy stability gives

$$t_1 Z_1 + (t_2 - t_1) Z_2 \stackrel{d}{=} t_2 Z.$$

Thus

$$U \longrightarrow U + t_1 Z_1 \longrightarrow U + t_1 Z_1 + (t_2 - t_1) Z_2$$

is a Markov chain, and data processing gives $F(t_2) \leq F(t_1)$. To see that $\lim_{t \rightarrow \infty} F(t) = 0$, fix u_0 in the support of U . The information-radius identity gives

$$\begin{aligned} F(t) &= \mathbb{E}_U D_{\text{KL}}(P_{U+tZ|U} \| P_{u_0+tZ}) - D_{\text{KL}}(P_{U+tZ} \| P_{u_0+tZ}) \\ &\leq \mathbb{E}_U D_{\text{KL}}(P_{U+tZ|U} \| P_{u_0+tZ}) \\ &= \mathbb{E} \log \left(\frac{1 + (Z + (U - u_0)/t)^2}{1 + Z^2} \right) \leq \frac{\mathbb{E}|U - u_0|}{t} \longrightarrow 0. \end{aligned}$$

The last inequality uses that $z \mapsto \log(1 + z^2)$ is 1-Lipschitz.

Since F is nonnegative, non-increasing, and convex, it must also hold that $\lim_{t \rightarrow \infty} F'(t) = 0$. Therefore

$$-F'(t) = \int_t^{+\infty} F''(s) ds,$$

and Eq. (58) follows from Eq. (57).

We now pass to general side information. Let $A := \text{supp}(U)$ and $m := |A|$. For a probability vector q on A , let $F_q(t)$ and $R_q(t)$ denote, respectively, the mutual information and the posterior-replica risk in the deterministic result with prior q . If $q_S(u) := \mathbb{P}(U = u | S)$, then

$$F(t) = \mathbb{E} F_{q_S}(t).$$

The deterministic result and the bound $R_q(t) \leq 1$ give, uniformly in q ,

$$0 \leq F_q(t) \leq \log m, \quad 0 \leq F_q''(t) \leq \frac{2}{t^2}, \quad 0 \leq -F_q'(t) \leq \frac{2}{t}.$$

Consequently, dominated convergence on every compact subinterval of $(0, \infty)$ yields

$$F'(t) = \mathbb{E} F_{q_S}'(t), \quad F''(t) = \mathbb{E} F_{q_S}''(t).$$

Moreover, $F_q(t) \rightarrow 0$ for every q , so dominated convergence also gives $F(t) \rightarrow 0$. Conditional on S , the replicas in the deterministic result have precisely the conditional law of two independent draws from $P_{U|S, U+tZ}$. Averaging the deterministic curvature bound therefore proves Eq. (57); the same convexity and endpoint argument proves Eq. (58). $\square$

A.2. The product Cauchy information–estimation inequality. The main result of this subsection immediately implies Proposition 4.1. Recall the replica cMMSE rcMMSE $_{\mu}(s)$ from Eq. (35).

Proposition A.2. *For every $\mu \in \mathcal{P}(T)$ and $t > 0$,*

$$-I'_{\mu}(t) \leq 4 \int_t^{\infty} \frac{\text{rcMMSE}_{\mu}(s)}{s^2} ds. \tag{66}$$

In particular,

$$I_\mu(t) \leq 4 \int_t^\infty \frac{\text{rcMMSE}_\mu(s)}{s} ds \leq 16 \int_t^\infty \frac{\text{cMMSE}_\mu(s)}{s} ds. \quad (67)$$

A.2.1. *A multi-scale Cauchy additive model and first steps.* To prove [Proposition A.2](#), we first consider a multiscale version of the Cauchy additive channel.

Let $X \sim \mu$, and let $Z_1, \dots, Z_n$ be independent standard Cauchy random variables, independent of X . For $\mathbf{t} = (t_1, \dots, t_n) \in [0, \infty)^n$, define

$$Y_{\mathbf{t}} := (X_i + t_i Z_i)_{i=1}^n, \quad I_\mu(\mathbf{t}) := I(X; Y_{\mathbf{t}}). \quad (68)$$

Thus $Y_{\mathbf{t},i} = X_i + t_i Z_i$, with $Y_{\mathbf{t},i} = X_i$ when $t_i = 0$. Derivatives of $I_\mu(\mathbf{t})$ below are taken only at points of $(0, \infty)^n$. The one-scale notation is consistent with this definition:

$$I_\mu(t) = I_\mu(t\mathbf{1}).$$

Fix $i \in [n]$. By the chain rule, if $Y_{\mathbf{t},-i}$ denotes $(Y_{\mathbf{t},j})_{j \neq i}$, then

$$\begin{aligned} I_\mu(\mathbf{t}) &= I(X; Y_{\mathbf{t},-i}) + I(X; Y_{\mathbf{t},i} \mid Y_{\mathbf{t},-i}) \\ &= I(X; Y_{\mathbf{t},-i}) + I(X_i; Y_{\mathbf{t},i} \mid Y_{\mathbf{t},-i}), \end{aligned} \quad (69)$$

where the last equality holds because $Y_{\mathbf{t},i}$ depends on X only through X_i . Notice that the first term does not depend on t_i , and therefore

$$\partial_{t_i} I_\mu(\mathbf{t}) = \partial_{t_i} I(X_i; Y_{\mathbf{t},i} \mid Y_{\mathbf{t},-i}).$$

For a scale vector $\mathbf{r}$ and a coordinate i with $r_i > 0$, let $X_{\mathbf{r}}^{(1)}, X_{\mathbf{r}}^{(2)}$ be conditionally independent draws from $P_{X|Y_{\mathbf{r}}}$, and define

$$\text{rcMMSE}_{i,\mu}(\mathbf{r}) := \mathbb{E} \left[1 \wedge \frac{|X_{\mathbf{r},i}^{(1)} - X_{\mathbf{r},i}^{(2)}|^2}{r_i^2} \right] \quad (70)$$

For $i \in [n]$ and $s > 0$, write

$$\mathbf{t}^{i,s} := (t_1, \dots, t_{i-1}, s, t_{i+1}, \dots, t_n).$$

In particular,

$$\text{rcMMSE}_{i,\mu}(s\mathbf{1}) = \text{rcMMSE}_{i,\mu}(s).$$

Applying [Lemma A.1](#) with $U = X_i$ and side information $S = Y_{\mathbf{t},-i}$, and using the independence of Z_i from (X_i, S) , gives

$$-\partial_{t_i} I_\mu(\mathbf{t}) = -\partial_{t_i} I(X_i; Y_{\mathbf{t},i} \mid Y_{\mathbf{t},-i}) \leq 2 \int_{t_i}^\infty \frac{\text{rcMMSE}_{i,\mu}(\mathbf{t}^{i,s})}{s^2} ds. \quad (71)$$

The following data-processing lemma compares $\text{rcMMSE}_{i,\mu}(\mathbf{t}^{i,s})$, for $\mathbf{t} = (t, \dots, t)$, with the one-scale quantity $\text{rcMMSE}_{i,\mu}(s)$ from [\(35\)](#).

Lemma A.3. *For every $0 \leq t \leq s$ and $i \in [n]$,*

$$\text{rcMMSE}_{i,\mu}((t, t, \dots, t)^{i,s}) \leq 2\text{rcMMSE}_{i,\mu}(s). \quad (72)$$

Proof. For squared loss, the claim follows from the usual data-processing inequality for MMSE and the stability of Cauchy noise; see, e.g., [WV11, Section V]. For the capped quadratic loss, we use a quadratic surrogate and apply data processing to that surrogate.

For $s > 0$, define

$$\psi_s(u, v) := \frac{|u - v|^2}{s^2 + |u - v|^2}, u, v \in \mathbb{R}. \quad (73)$$

For all $u, v \in \mathbb{R}$,

$$\frac{1}{2} \left(1 \wedge \frac{|u - v|^2}{s^2} \right) \leq \psi_s(u, v) \leq 1 \wedge \frac{|u - v|^2}{s^2}. \quad (74)$$

Moreover, for any $s > 0$, the Cauchy kernel

$$k_s(u, v) := 1 - \psi_s(u, v) = \frac{s^2}{s^2 + (u - v)^2} \quad (75)$$

is positive definite. Indeed, it can be written as the Fourier transform of a Laplace distribution, so Bochner's theorem [Boc59] implies the result:

$$k_s(u, v) = \int_{\mathbb{R}} e^{i\xi(u-v)} \nu_s(d\xi), \quad \nu_s(d\xi) := \frac{s}{2} e^{-s|\xi|} d\xi. \quad (76)$$

Thus, for any $s > 0$, the Moore–Aronszajn theorem [Aro50] yields a real Hilbert-space feature map Γ_s satisfying, for all $u, v \in \mathbb{R}$,

$$\|\Gamma_s(u)\| = 1, \quad \langle \Gamma_s(u), \Gamma_s(v) \rangle = k_s(u, v),$$

and

$$\psi_s(u, v) = \frac{1}{2} \|\Gamma_s(u) - \Gamma_s(v)\|^2. \quad (77)$$

The quadratic structure gives the following useful property. For any channel with signal U , if $U^{(1)}, U^{(2)}$ are i.i.d. posterior samples of U given an observation vector $\mathcal{O}$, then the Nishimori identity from Section 2.1 gives

$$\mathbb{E}[\psi_s(U^{(1)}, U^{(2)})] = \frac{1}{2} \mathbb{E} \|\Gamma_s(U^{(1)}) - \Gamma_s(U^{(2)})\|^2 = \mathbb{E} \|\Gamma_s(U) - \mathbb{E}[\Gamma_s(U) | \mathcal{O}]\|^2. \quad (78)$$

Therefore $\mathbb{E}[\psi_s(U^{(1)}, U^{(2)})]$ can only decrease when the observation becomes more informative. Indeed, suppose $U \rightarrow \mathcal{O}' \rightarrow \mathcal{O}$ is a Markov chain, and set

$$M' := \mathbb{E}[\Gamma_s(U) | \mathcal{O}'], \quad M := \mathbb{E}[\Gamma_s(U) | \mathcal{O}].$$

The Markov property gives $\mathbb{E}[\Gamma_s(U) | \mathcal{O}', \mathcal{O}] = M'$, and hence

$$\mathbb{E} \langle \Gamma_s(U) - M', M' - M \rangle = 0.$$

Since $\Gamma_s(U) - M = (\Gamma_s(U) - M') + (M' - M)$, the Pythagorean identity gives

$$\mathbb{E} \|\Gamma_s(U) - M\|^2 = \mathbb{E} \|\Gamma_s(U) - M'\|^2 + \mathbb{E} \|M' - M\|^2.$$

Thus the posterior feature variance for $\mathcal{O}'$ is no larger than that for $\mathcal{O}$; see Blackwell [Bla53].

Now specialize to the diagonal scale $\mathbf{t} = (t, \dots, t)$ and take $s \geq t$. From the mixed observation

$$Y_{(t, \dots, t, s, t, \dots, t)}$$

we can generate the all- s observation $Y_s = X + sZ$ by adding, to every coordinate $j \neq i$, an independent Cauchy noise of scale $s - t$. Indeed, for independent standard Cauchy variables ξ, ξ' and $0 \leq t \leq s$, stability gives

$$t\xi + (s - t)\xi' \stackrel{d}{=} s\xi. \quad (79)$$

Hence the mixed observation is more informative than the all- s observation. By Eqs. (74) and (78), we can then conclude

$$\text{rcMMSE}_{i,\mu}((t, \dots, t)^{i,s}) \leq 2\text{rcMMSE}_{i,\mu}(s). \quad (80)$$

□

Proof of Proposition A.2. Notice that using Eqs. (71) and (72) we have

$$-I'_\mu(t) = -\sum_{i=1}^n \partial_{t_i} I_\mu(t, \dots, t) \leq 4 \sum_{i=1}^n \int_t^\infty \frac{\text{rcMMSE}_{i,\mu}(s)}{s^2} ds = 4 \int_t^\infty \frac{\text{rcMMSE}_\mu(s)}{s^2} ds. \quad (81)$$

Fix $x_0 \in T$. The information-radius identity, followed by tensorization of relative entropy for the product Cauchy channel, gives

$$\begin{aligned} I_\mu(t) &\leq \mathbb{E}_X D_{\text{KL}}(P_{X+tZ|X} \| P_{x_0+tZ}) \\ &= \sum_{i=1}^n \mathbb{E}_X D_{\text{KL}}(P_{X_i+tZ_i|X_i} \| P_{x_{0,i}+tZ_i}) \\ &\leq \frac{\mathbb{E}\|X - x_0\|_1}{t} \longrightarrow 0. \end{aligned}$$

Here the last inequality is the shifted-Cauchy estimate used in the proof of Lemma A.1. Thus Eq. (81) and Tonelli's theorem give

$$I_\mu(t) = \int_t^\infty [-I'_\mu(u)] du \leq 4 \int_t^\infty \int_u^\infty \frac{\text{rcMMSE}_\mu(s)}{s^2} ds du \quad (82)$$

$$= 4 \int_t^\infty \frac{s-t}{s^2} \text{rcMMSE}_\mu(s) ds \leq 4 \int_t^\infty \frac{\text{rcMMSE}_\mu(s)}{s} ds. \quad (83)$$

Using Lemma 4.2 concludes the proof. □

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